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VISUAL PERCEPTION

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ABSTRACT

In order to understand the mechanisms of visual perception, we start with a short presentation on the basic elements of the eye which are involved in the visual perceiving. The article talks about three main topics. In the first part, we introduce the bio-chemical model which describes how the retina transforms the light stimulation into an electrical pulse. Secondly, we show how interactions between photoreceptor cells, mediated by connections in the nervous system, can lead to information processing such as making distinction between two nuances. The third topic is color vision, where the relation between spectrum and color perception is discussed.

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1. INTRODUCTION

Within the frame of recent development of optoelectronic devices for image capturing which is accompanied by many technological challenges, the structure and function of the human image capturing device – the human eye – became even more astonishing. While comparing the resolution, sensitivity and dynamical range of the human eye to those of recent digital cameras, one can conclude that the eye is still supreme over these optoelectronic devices. It is even more amazing when one thinks about the efficiency of the information flow from the eye to our brain that should actually translate the moving image to a flow of associations in real time, which at least for the current devices seems as a science fiction. Therefore, it is very important to understand the basic biophysical principles that govern the function of the human eye [1].

Ongoing research has identified a long list of biological molecules and structures involved in neural signalling. Yet, we do not understand what most of this biological circuitry does. To this end, the function of vertebrate retina needs to be explored ([2], [3], [4]).

2. STRUCTURE OF THE EYE

The eyeball consists of a lens (cornea, crystalline lens) which refracts the light, and a detector (retina) on which the rays are focused. The basic components are shown on *figure 1*.

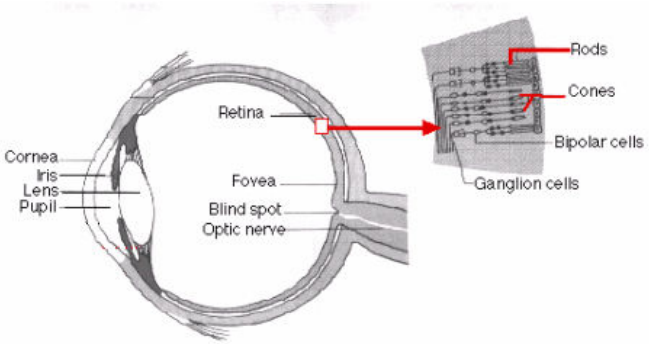


Figure 1: Eye scheme [9]

The cornea is a transparent part of the outer coat of the eyeball. This is the first and most powerful lens of the optical system of the eye. The highest dioptre of all the optical parts of the human eye comes from the fact that the light experiences great change in refraction index

when passing from air into the curved cornea. The refraction index change is for example much smaller when the light passes from the pupil to the human lens and form the lens to the vitreous body.



Figure 2: Scanning electron micrograph of the rods and cones of the primate retina [8]

The **retina** represents the light detector of the eye, constructed from light-sensitive cells, **rods** and **cones**, pigment cells and neuronal cells that continue into the **optic nerve** which conducts visual impulses to the brain. The small area on the retina that provides our most central acute (and color) vision is the **fovea**.

Rods (the yellow cells on *figure 2*) are specialized for detecting brightness, while **cones** (the short red cells on *figure 2*) perceive color. There are three kinds of cones: the ones that absorb long-wavelength light (red), the ones that absorb middle-wavelength light (green) and the ones that absorb short-wavelength light (blue).

3. FROM PHOTON TO ELECTRIC PULSE

When the light falls on the retina, receptors are stimulated and an electric signal is produced. In this section, I will talk about how the photon generates an electric pulse.

3.1 Retina in the dark

All the photon absorption in the rods and cones actually occurs in their membranes which contain channels for the passive transport of ions (red and rose object on *figure 3a*), pumps for the active transport of ions (orange object on *figure 3a*) and proteins that can capture the photons. When the retina is in the dark, the sodium and potassium channels for passive transport are opened. Because of the active transport, which pushes 3 Na^+ ions outside, while it lets only 2 K^+ ions inside, there are more positively charged ions outside than inside the cells [9]. The membrane is therefore negatively polarized and it maintains the potential of -65 mV (*figure 3b*).

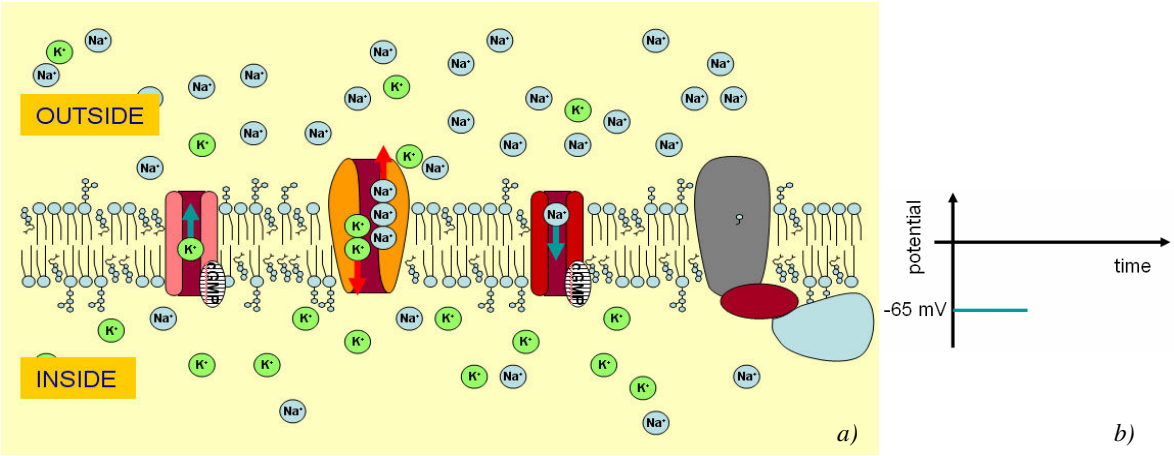


Figure 3: a) Receptor membrane in the dark: Na^+ and K^+ channels are opened. The retinal is not activated yet. b) Potential of the receptor membrane in the dark (right)

3.2 Retina in the light

Let us now have a look at what happens when a photon reaches the retina [9].

Special proteins for absorbing light (gray object on the right of *figures 3* and *5*) called **opsin** are coupled to a set of molecules called **retinal.** The latest consists of many double bonds which can be excited after absorbing visual photons.

When the photon reaches the retinal, it changes its conformation. Cis-retinal changes into trans-retinal (*figure 4*).

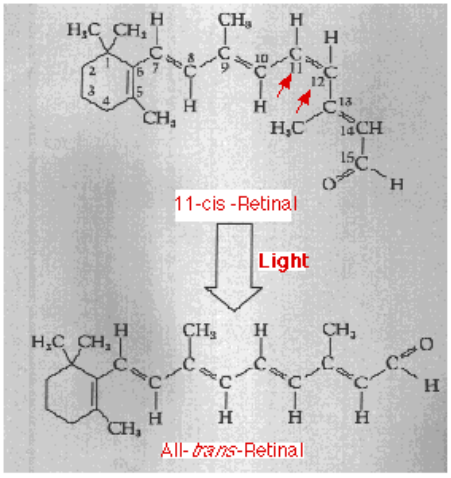


Figure 4: Cis-trans isomerisation of retinal [9]

Then, a chain of bio-chemical processes happens. The formation of **trans-retinal** activates **opsin**, conformational change of opsin activates a protein complex called **transducin** which

activates an enzyme that breaks down **cyclic GMP** (Cyclic guanosine monophosphate). Lower concentration of cGMP closes Na^+ and K^+ channels (the rose and the red objects on *figure 5a*). While the channels of passive transport are closed, the sodium/potassium pump (the orange object on *figure 5a*) keeps on pushing only two potassium ions (K^+) into the cell for every three sodium ions (Na^+) it pumps outside, thus the positive charge is going out. The interior part of the cell membrane becomes still more negative, as shown on *figure 5b*. This way, we get a local perturbation which travels along the membrane till it reaches the synapse with a neural cell [9].

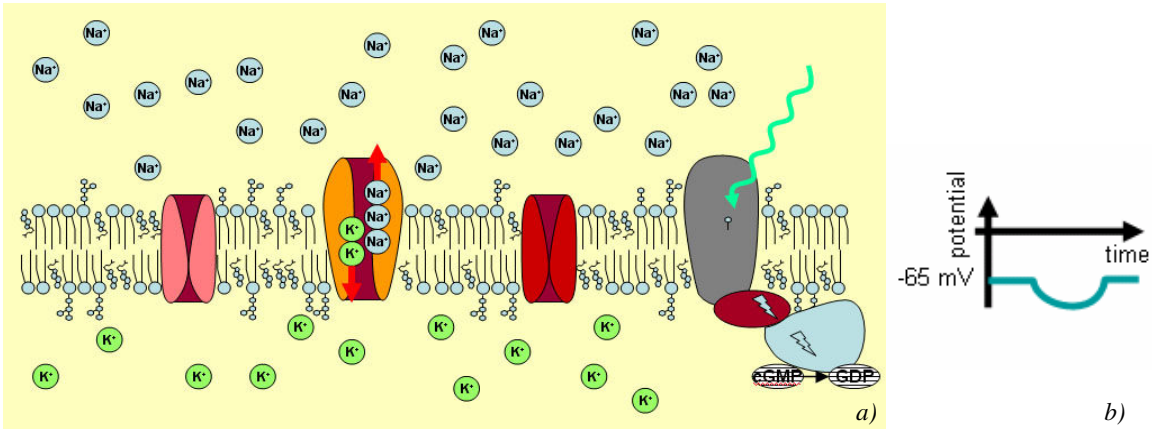


Figure 5: a) The excited receptor membrane: a photon activates retinal, the bio-chemical process goes off, Na^+ and K^+ channels are opened and the membrane becomes hyperpolarized. b) Potential of the excited receptor membrane (right).

3.3 Can a human see a single photon?

The human eye is very sensitive but can we see a single photon? The answer is that the sensors in the retina can respond to a single photon. One photon hyperpolarizes the membrane by *1 mV*. However, neural filters only allow a signal to pass to the brain to trigger a conscious response when at least about five to nine arrive within less than *100 ms*. If we could consciously see single photons, we would experience too much visual "noise" in very low light, so this filter is a necessary adaptation, not a weakness [6].

3.4 From the receptor to the nerve

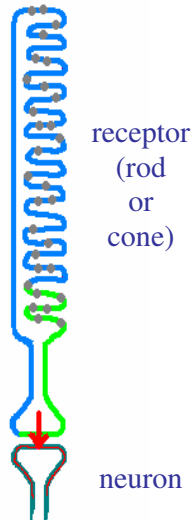


Figure 6: Release of neurotransmitters at the synapse of the detector cell

Perturbation on the receptor membrane causes the release of the neurotransmitters (*figure 6*) which cause the polarization of the neuron's membrane (*figure 7*).

When the neuron is not excited, the channels for passive transport are closed (*figure 7a*). Then, the neurotransmitter opens the Na^+ channel (*figure 7b*) so the Na^+ ions go inside the cell and the potential becomes positive. The positive potential opens the K^+ channels which can consequently go out of the cell (*figure 7c*) [9]. The potential becomes negative again but with the opposite distribution of ions, which is finally equilibrated by Na/K pump. This way, we get an amplified pulse (*figure 7d*) which is the input in the neural network[2].

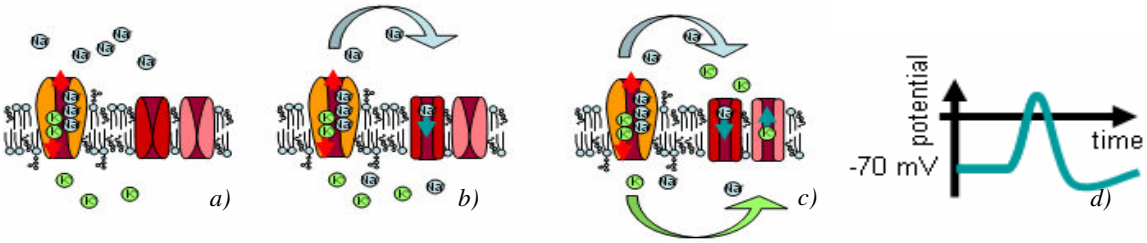


Figure 7: Hyperpolarization of ganglion cell: a) Na^+ and K^+ channels are closed; b) The neurotransmitter opens the Na+ channel; c) Na^+ and K^+ channels are opened; d) Potential on the neuron membrane.

The grater the light-intensity the shorter is the time between two photons absorbing on the receptor and the grater is the rate of firing. This means that the light intensity is frequency-coded. The basic unit of information transfer is the action potential, an electrical transient of about *1 ms* duration, and *100 mV* amplitude propagating with speeds ranging from *1* to *100m/s* and impulse frequencies up to approximately *1000* per second [1].

Before the signal is transduced further to the optical nerve, it passes a complex neural network in the retina. It consists of a complex array of bipolar cells, ganglion cells and other interneurons. Together they form a neural network [2] that translates the signal from the rods

and cones to the brain. As this neural network possesses very interesting properties, it will be discussed in the next section.

4. CONTRASTING NUANCES

In an eye, there are *130 million* receptors but only *1 million* nerve fibers transduce the optical information to the brain. Where does the information go?

The retina compresses the information flow in order to increase the quality of the information and to provide some basic geometrical description already at the level of detector even before the information flow reaches the neural networks of the brain. The input into the brain is therefore not like a BMP and the retina is not simply a sheet of photocells. Much more complex processes are carried out before the signal is sent to the brain. Let us have a closer look at one of these processes.

4.1 Contrasting the border

An eye is not very efficient in distinguishing between two shades but it is very effective in contrasting the border region between two very similar areas.

The figure below is formed of bands with uniform brightness that gradually increases from left to right. Even though each band is completely uniform, it gives an impression that the left edge is brighter than the right one. This is the effect of contrasting on the border.

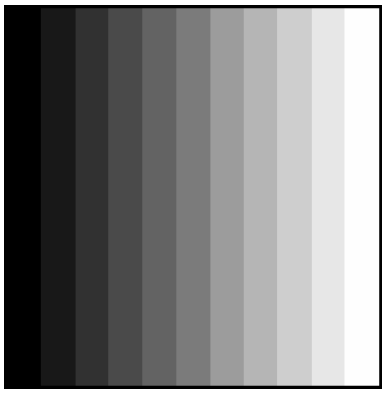


Figure 9: uniform bands which appear to have intensity variations that enhance the contrast at the border

The distinction between two similar shades is enabled due to the inhibitory and excitatory connections between the neurons in the retina. The pattern of illumination on the retina is

transformed into a pattern of perceived brightness, which is contrasted on the border between two shades.

4.2 Mathematical model

Let us consider an array of neurons that are laterally connected (*figure 10*) [1]. The properties of a lateral inhibition network can be described mathematically by the following equation

$$I_i(t) = I_i^0(t) \pm \lambda \sum_k \int K_{ik}(t-t')I_k(t')dt' \qquad (1)$$

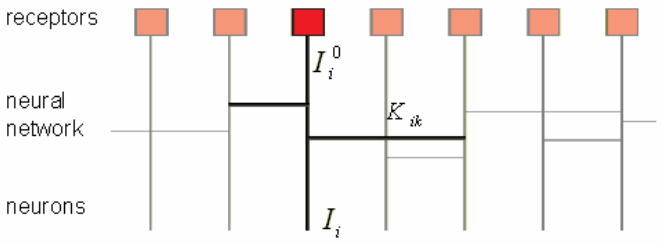


Figure 10: Schematic diagram of an array of photoreceptors, showing lateral connections between neighboring neurons.

As it is explained in the section 4.3 the light intensity is frequency-coded. Therefore, $I_i^0(t)$ represents the frequency of firing at the entrance of the neural network (the input on the *i*-th neuron caused by the light falling on the receptor).

$I_i(t)$ is the enhanced/reduced frequency of firing (the output of the *i*-th neuron in the array), and K_{ik} represents the effect of synaptic connection between the *i*-th neuron and its neighbors. λ is a constant that represents the strength of the synaptic inhibition/excitation relative to the magnitude of the direct excitation. The sign is negative for the inhibitory connections and positive for the excitatory connections.

If we consider that the illumination is steady, we can neglect the time dependency. In this case, the problem is reduced to a linear equation [1].

$$I_i = I_i^0 \pm \lambda \sum_k K_{ik} I_k \qquad (2)$$

This system of linear equations is used in the next section for making a model that will help us understand why a uniform area might appear to have brightness variations as on *figure 9*.

4.3 A simple numerical model

Here is proposed a simple model that describes how we perceive brightness in the case of steady illumination by an edge of light. Let us take that we have an infinite dark area on the left and an infinite bright area on the right (*figure 12a*). We will explore how inhibitory and excitatory connections influence the nuance that is finally "seen".

In the retina, the receptors are placed in a plan but in our case a one-dimensional model is good enough. We consider a linear array of receptors: the left ones are in the dark and the right ones in the light.

A neuron sends out dendrites by a random walk, thus the probability that the *i*-th neuron is connected to the *k*-th neuron is greater the greater the distance d_{ik} is. A good assumption for this distribution is a Gaussian fall off

$$K_{ik} \propto \exp\left(-\frac{d_{ik}^{\,2}}{\sigma^2}\right), \qquad (3)$$

where d_{ik} represents the dimensionless distance between two cells (counting cells along the retina) and σ rapresents the variance of the Gaussian distribution.

This model is just an illustration so the parameters λ and σ are simply invented. The strength of synaptic inhibition / excitation is taken to be $\lambda = 0.1$ and $\sigma = 3$. Another simplification is that only connections between the first 6 neighboring neurons in a grid are taken into consideration. By solving the system of equations (2) we come to an interesting result, which associates to apparent brightness variations that can be observed on *figure 9*.

Let us consider only inhibitory connections (green pattern on *figure 11*). We see that cells on the far right of the edge are brightly lit (gray pattern on *figure 11*) and would be firing maximally except for inhibition by their neighbors which are also brightly illuminated. Thus, the firing frequency is below the maximum for isolated cells. Cells on the right, but closer to the boundary, are less inhibited because some of their neighbors are in the dark. Their outputs are at higher frequency. Cells on the left side of the boundary are in the dark but are also inhibited by their neighbors in the light, so that firing frequency is depressed. Cells on the far left are almost not inhibited. This way, we get an enhanced contrast on the border. Some

animals really have neurons that make only inhibitory connections, for example in the eye of the horseshoe crab, *Limulus* [1].

In general, neurons in the retina make both excitatory and inhibitory connections, which prevents a considerable decrease of brightness of the white area. The consequence is that the perceived brightness near the border oscillates (red pattern on *figure 11*).

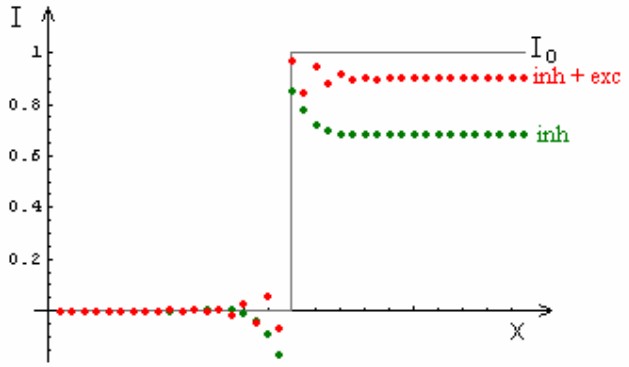
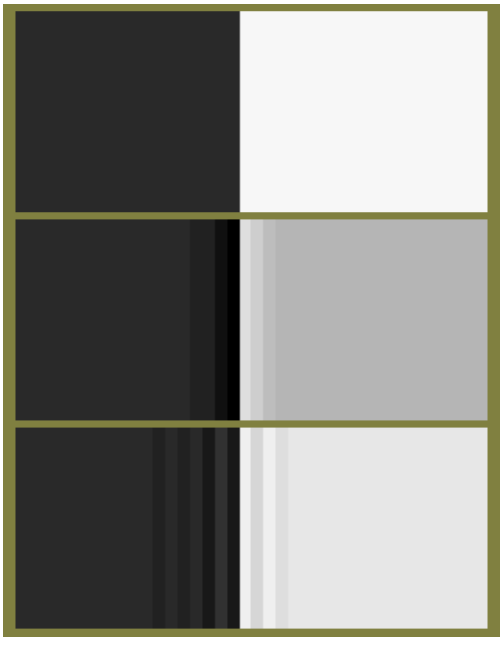


Figure11: Effect of inhibition (green pattern) and inhibition + excitation (red pattern) in the retina. I_0 represents the excitatory input.



We can see that lateral inhibition enhances the contrast at the edge but the perceived brightness away from the edge is reduced (*figure 12b*). By adding some excitatory connections the perceived brightness is less diminished and the contrast on the edge is still enhanced (*figure 12c*). Note that some bands of intensity variations are created. This explains the famous psychophysical phenomenon of Mach bands [10]. (*Mach Bands - The optical illusion that enhances contrast between regions of varying intensity by creating dark and light lines at the border of the contrasting areas.*)

Figure 12:
a) Uniform, steady illumination by an edge of light
b) Model of perceived pattern considering inhibitory connections.
c) Model of perceived pattern considering inhibitory and excitatory connections.

This model is a linear network in which the frequency of an individual neuron firing is linearly dependent on the firing frequencies of its neighbors. Already in a very simple model that considers only 6 neighboring cells, we can explain some basic properties of this neural network such as perception of brightness variation across the border between dark and bright regions. However, several properties have not been considered, as time dependence and even nonlinear relation between input and output.

5. COLOR VISION

"…color means pigment to an artist, an internal perception to a psychologist, the response of a neural system to a physiologist, an aspect of radiant energy to a physicist, and a property of objects and lights to a man in the street." Danger [10]

5.1 Cone absorbance and sensation of color

The sensation of color arises primarily from different absorptions of the receptors. Cones are mainly responsible for color vision while rods are not sensitive to color. The three classes of color detectors have been corroborated by spectrophotometric measurements on single cones, for example in the goldfish retina [4]. Three kinds of cones correspond to the three different absorption maximum, as shown in *figure 12*. Thus, each class of cone absorbs photons in a band about its maximum (R:575 nm; G:535 nm; B:445 nm). The electrical output of each cone depends upon the number of photons absorbed, and the sensation of color comes about from the particular mixture of neural outputs originating from each cone.

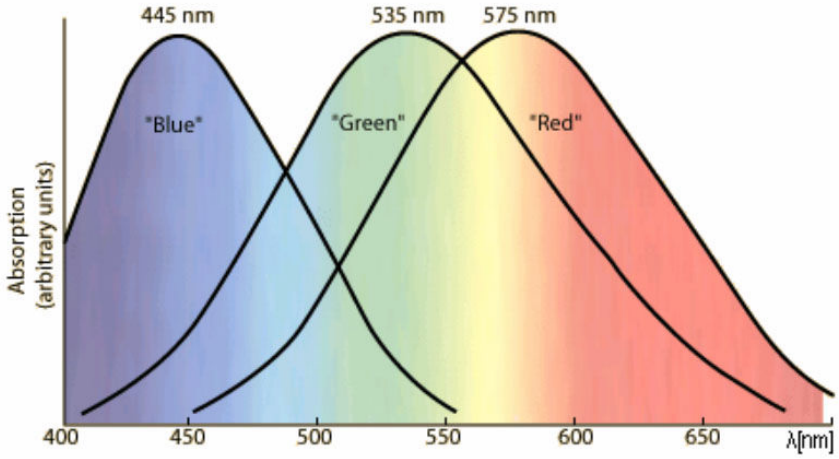


Figure 13: Approximate absorption spectra for "red", "green" and "blue" cones.[7]

E.g.: If mainly the "red" cones are excited, the eye perceives red, if mainly the "green" cones are excited, the eye perceives green, and if mainly the "blue" cones are excited, the eye perceives blue. Other colors are obtained if two or three different types of cones are excited simultaneously. If blue and green cones are excited, the impression of color is cyan. If blue and red cones are excited, the impression of color is magenta. If red and green cones are excited, the impression of color is yellow.

Note that heights of the curves on *figure 13* are in arbitrary units. In fact, the three types of cones have different sensitivity (different absolute heights of absorption curves) but this is not directly related to the perception of the color. Before reaching the brain, the signal is still filtered by the neural network [3]. Thus, from the absorption functions we cannot find out which color matches a certain mixture of lights. What we need to connect the spectra of the light with the resulting color are the color matching functions (section 5.4).

5.2 Spectral power distribution

Light may be precisely characterized by giving the power of the light at each wavelength in the visible spectrum. The resulting spectral power distribution (SPD) contains all the basic physical data about the light. It serves as the starting point for quantitative analyses of color. An example of SPD of light coming from common objects is shown on *figure 14*.

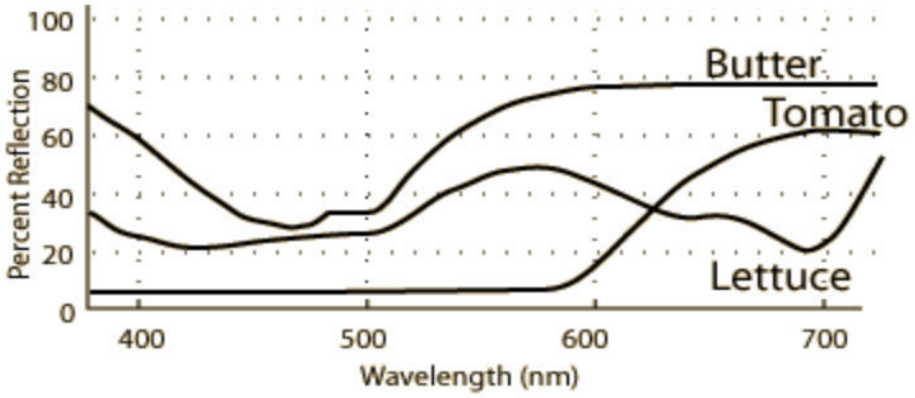


Figure14: An example of the spectral power distribution of light from common objects [7]

From here, we can predict the color that is perceived by the human eye but the opposite proceeding is less accurate. The eye perceives different inputs as the same color [7]. E.g.: The mixture of green and red light (in right proportions) stimulates the receptors in the same way as the yellow light does, so one sees yellow in both cases.

5.3 CIE chromaticity diagram

All the colors that can be perceived by the human eye are represented in the CIE chromaticity diagram *(figure 15)*. Along the outside curve of the chromaticity diagram we have the rainbow spectrum of pure spectral colors. We know that the spectral colors can be one-to-one correlated with light wavelength. The bottom line represents the purples. They can only be produced by mixing red and blue lights. [7]

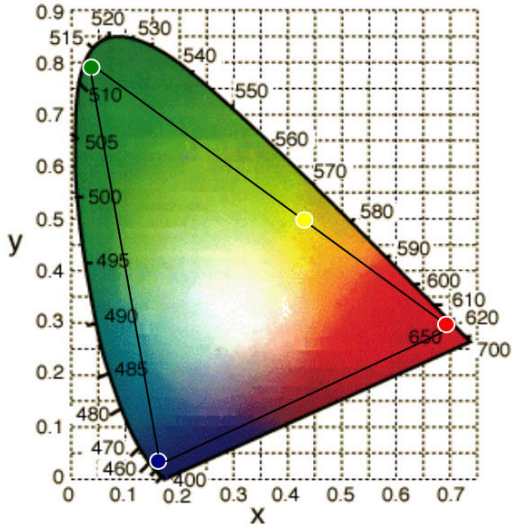


Figure 15: The CIE chromaticity diagram [7]

A useful propriety of the chromaticity diagram is that we can produce any color that lies between two different colors by mixing them in right proportions. Eg: by mixing green and blue lights we can get yellow light *(figure 15)*.Thus, by mixing three different lights we can produce any color that lies in the triangle between them. **By mixing red, green and blue lights we can cover a great part of visible colors,** so these three colors are called "the primary colors".

Note that we are talking here about additive color mixing. It should not be confused with the subtractive color mixing, which corresponds to mixing pigments. We will never obtain a yellow by mixing green and red paint.

5.4 Matching functions

While the spectrum of the light coming from an object is well defined, it is not trivial to define the color. In fact, there are different systems describing the color [5]. One of the possibilities is to use three primary colors like blue, green and red as vectors with which to map color space. We can represent the color with 3 coordinates *a*, *b* and *c*:

$$C = aR + bG + cB\,, \qquad (4)$$

where *R, G* and *B* represent the primary colors red, green and blue. *a, b* and *c* are the intensities of the primary colors. Later in this section, we will try to find out what is the relation between the spectrum and the matching *a, b* and *c* coordinates.

In fact, the relation between the SPD of the falling light and the resulting color sensation is rather complex and the subjectivity of human factor cannot be avoided as the color sensation can be measured by asking a human subject what color appears to match a spectral mixture of lights [7].

By making several experiments, color matching functions were defined (*figure 16*). If the spectrum of a color is weighed by these functions, the coordinates *a, b* and *c* can be calculated. E.g.: to calculate the coordinate *a* we multiply the function b_λ (it is drawn on *figure 16*) and the spectrum. Then, we integrate it to get a value. We do analogue for *b* and *c* for green and red color intensity.

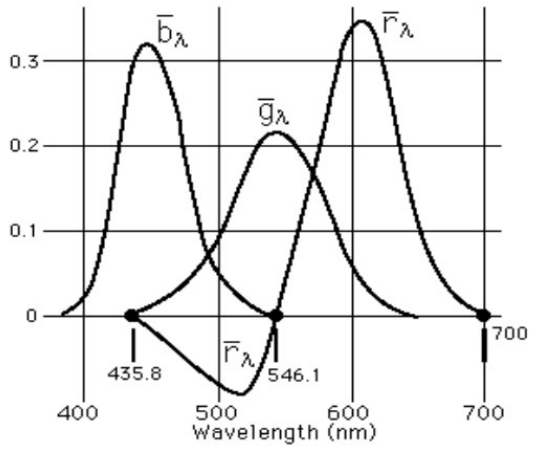


Figure 16: The RGB matching functions [7]

5.6 Configuration of the receptor cells

Even though we might not even notice it in our daily life, the color perception is limited only to the central vision. The peripheral vision is better for perceiving brightness variation and movement. This is a consequence of cones and rods distribution, as it will be explained further on.

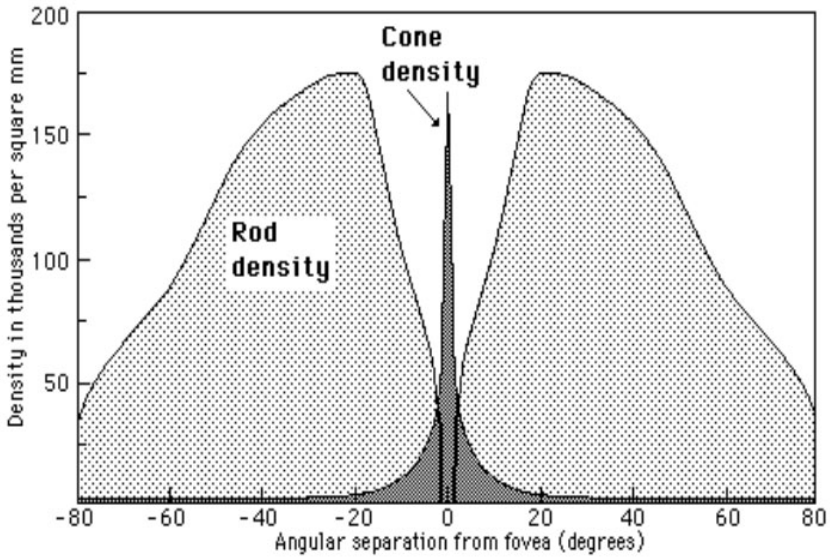


Figure 17: Rod and cone density on the retina [7]

As we can notice on *figure 17*, the cones are very concentrated in the central part (the fovea), and there are almost none of them in the periphery [7]. This is why we cannot distinguish colors with the peripheral vision.

There are many rods (120 million) so they can cover wide angle vision (*figure 17*). Because of their numerousness they make a lot of neural interconnections, therefore they are more adapted to contrasting and detecting movement than cones. Their role is to perceive brightness, so they have to be very sensitive to light. If we drew the absorption spectrum, the peak would be much higher compared to the cones. Cones are much less light sensitive and this is why we do not see colors at twilight.

There are fewer cones in the retina – only 6 to 7 millions. They are concentrated in the central yellow spot known as the macula. The central part of the macula is the fovea, where there are no rods. The ratio between "green" cones, "red" cones and "blue" cones is 64 : 32 : 2.

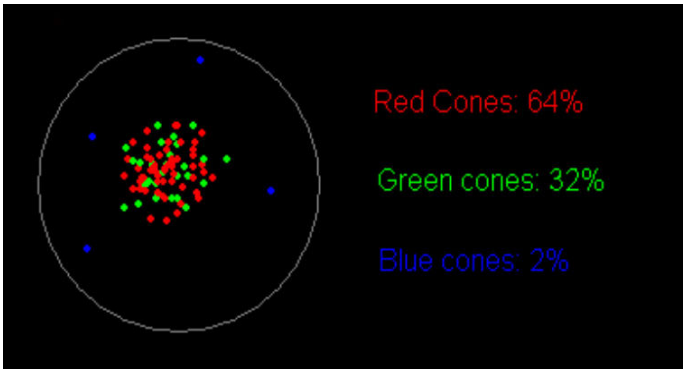


Figure 18: Cone configuration

Note that the "blue" cones are less numerous and mostly found outside the fovea (*figure 18*). The reason is that the refractive index for blue light is enough different from red and green making the blue image slightly out of focus when red/green image is in focus (due to chromatic aberration) demanding lower angular resolution for blue image. The consequence of the lower number of the "blue" cones is the apparent lower intensity of the perceived blue colors in comparison with green and red color [7].

5.7 Color at twilight

At low light levels, we do not distinguish color. Here, the rods play a more important role, as they are more light sensitive. When the ambient light level changes, there are apparent shifts of color.

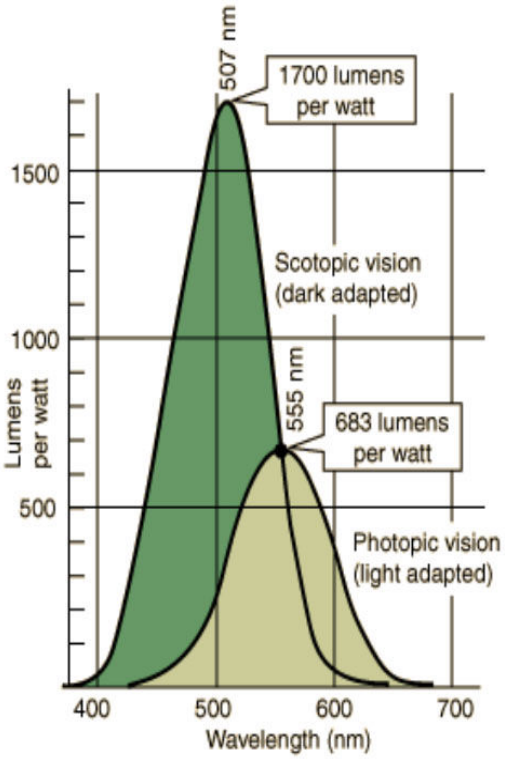


Figure 19: The spectral luminous efficacy for human vision [7]

The eye has night and day adapted vision [7]. Scotopic vision is adaptation to the dark-light conditions. On *figure 19*, we see the spectrum of luminous efficacy for both day and night vision. We notice that night-adapted vision peaks at shorter wavelengths (λ=507nm→green) and it has grater sensitivity.

The lumen is defined such that the peak of the photopic vision curve has a luminous efficacy of 683 lumens/watt. From here we get

$$\frac{F[lum]}{P[W]} = eff \times 683 lum/W, \quad (5)$$

where F represents luminous flux in lumens, P represents radiant power in watts and *eff* represents efficacy [7].

The scotopic vision is more sensible to green lights comparing to lights with other wavelengths. This explains why the petals of a red rose are much brighter than its leaves by day, but the leaves are brighter at twilight [7].

To sum up, due to the physiology of the eye which has 3 types of color receptors, we can define color with 3 coordinates. The relation between the spectrum of incoming light and perceived color is modeled with matching functions. To determine them, we cannot avoid the influence of subjectivity because each person perceives color slightly differently. As nearly all the cones are concentrated in the fovea, we cannot see color with the peripheral vision. Moreover, the color perception is diminished and changed at twilight when the rod-absorbance greatly influences color vision. In this case, the color matching functions are not reliable anymore.

6. CONCLUSION

The eye is an amazing organ full of secrets and one can learn a lot from it. One can understand its function from basic physical principles such as transforming a photon into an electrical pulse, contrasting and perceiving the color. Each of these topics makes people feel like standing on the edge of an immensity of things they do not know. In a way, thinking about vision seems like thinking about the laws of nature. The more you understand the more you realize that you actually do not understand.

7. REFERENCES

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